VMTS: Vision-Assisted Teacher-Student Reinforcement Learning for Multi-Terrain Locomotion in Bipedal Robots

Fu Chen, Rui Wan, Peidong Liu, Nanxing Zheng and Bo Zhou*

***Abstract*— Bipedal robots, due to their anthropomorphic design, offer substantial potential across various applications, yet their control is hindered by the complexity of their structure. Currently, most research focuses on proprioception-based methods, which lack the capability to overcome complex terrain. While visual perception is vital for operation in human-centric environments, its integration complicates control further. Recent reinforcement learning (RL) approaches have shown promise in enhancing legged robot locomotion, particularly with proprioception-based methods. However, terrain adaptability, especially for bipedal robots, remains a significant challenge, with most research focusing on flat-terrain scenarios. In this paper, we introduce a novel mixture of experts teacher-student network RL strategy, which enhances the performance of teacher-student policies based on visual inputs through a simple yet effective approach. Our method combines terrain selection strategies with the teacher policy, resulting in superior performance compared to traditional models. Additionally, we introduce an alignment loss between the teacher and student networks, rather than enforcing strict similarity, to improve the student's ability to navigate diverse terrains. We validate our approach experimentally on the Limx Dynamic P1 bipedal robot, demonstrating its feasibility and robustness across multiple terrain types.**

Index Terms—Bipedal robots, Reinforcement learning, Control for visual perception

Fig. 1. We have developed an end-to-end vision-based reinforcement learning control strategy for bipedal robots, enabling them to overcome various terrains, such as 15 cm height differences, 30-degree slopes, and grasslands.

I. INTRODUCTION

Bipedal robots, owing to their anthropomorphic structure, offer significant potential and value across various application domains. However, the inherent complexity of their design presents a persistent challenge in control. Moreover, to effectively function within human-centric environments, visual perception plays a pivotal role. Yet, the integration of visual input further exacerbates the difficulty of the control task.

Over the past few decades, numerous researchers have proposed a variety of advanced methods in this field [1]–[4]. Among the most prominent is Boston Dynamics' Atlas robot, which is capable of performing a wide range of complex movements [5]–[7]. However, most of these methods rely on the integration of odometry to generate online maps and require substantial time for offline planning, often resulting in limited generalization performance.

Recently, reinforcement learning has yielded promising results in the control of legged robots, demonstrating significant advancements in the field [8]–[13]. Currently, research on reinforcement learning for legged robots primarily focuses on proprioception. Lee et al. [8] were among the first to propose a teacher-student network learning approach on legged robot. In the teacher-student network, the teacher policy, endowed with privileged information, undergoes training in the first phase. During the second phase, the student policy, which relies solely on proprioceptive data, imitates the actions of the teacher policy. This methodology allows the student policy to achieve performance closely resembling that of the teacher policy while only utilizing the necessary information, making it a widely studied approach [10], [14], [15]. Different to the teacher-student two-phase training approach, there are also numerous methods that require only a single-phase training process. DreamWaQ [9] proposed the use of Variational Autoencoders (VAE) to estimate implicit representations of states. Wang et al. [16] applied this estimation method to bipedal robots and evaluated the impact of the estimated states on the locomotion performance of the robot. Fu et al. [17] proposed incorporating privileged learning of encoded proprioceptive information during the training process, and applied this approach to mobile manipulation tasks. Wang et al. [18] proposed a method for simultaneously training both teacher and student networks, which significantly enhances the performance of the student policy with minimal impact on the teacher policy. However, in the absence of external perception, overcoming terrain remains a significant challenge for legged robots, particularly bipedal robots. As a result, current research on bipedal robots primarily focuses on flat-terrain environments.

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VMTS：面向双足机器人多地形运动的视觉辅助师生强化学习

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摘要——**双足机器人凭借其仿人设计，在各类应用中展现出巨大潜力，但其控制因结构复杂性而受到制约。目前，大多数研究聚焦于基于本体感知的方法，这些方法缺乏应对复杂地形的能力。尽管视觉感知对于在人类中心环境中运行至关重要，但其引入进一步增加了控制的复杂性。最近的强化学习方法在提升腿式机器人运动方面展现出前景，尤其是基于本体感知的方法。然而，地形适应性（尤其是针对双足机器人）仍是一项重大挑战，大多数研究仅关注平坦地形场景。本文提出了一种新颖的专家混合师生网络强化学习策略，通过一种简单而有效的方法，增强了基于视觉输入的师生策略的性能。我们的方法将地形选择策略与教师策略相结合，相比传统模型展现出更优越的性能。此外，我们在师生网络之间引入了对齐损失，而非强制要求严格相似性，以提升学生在多样化地形中导航的能力。我们通过在Limx Dynamic P1双足机器人上进行实验验证了该方法，证明了其在多种地形类型上的可行性和鲁棒性。**

图1. 我们开发了一种端到端基于视觉的强化学习控制策略，使双足机器人能够克服多种地形，例如15厘米高度差、30度斜坡和草地。

索引术语——双足机器人，强化学习，视觉感知控制

I. 引言

双足机器人因其类人结构，在各个应用领域展现出巨大的潜力和价值。然而，其设计固有的复杂性给控制带来了持续的挑战。此外，为了在人类环境中有效运作，视觉感知起着关键作用。然而，视觉输入的整合进一步加剧了控制任务的难度。

在过去的几十年里，众多研究人员在该领域提出了各种先进方法 [1]–[4]。其中最著名的是波士顿动力的Atlas机器人，它能够执行各种复杂的动作 [5]–[7]。然而，这些方法大多依赖于里程计集成来生成在线地图，并且需要大量的时间进行离线规划，通常导致泛化性能有限。

近年来，强化学习在腿式机器人控制方面取得了令人瞩目的成果，在该领域展示了显著的进步 [8]–[13]。目前，研究关于腿式机器人的强化学习主要聚焦于本体感觉。Lee等人 [8] 是最早提出将师生网络学习方法应用于腿式机器人的研究者之一。在师生网络中，拥有特权信息的教师策略在第一阶段进行训练。在第二阶段，仅依赖本体感觉数据的学生策略会模仿教师策略的动作。这种方法使学生策略在仅利用必要信息的情况下，能够达到与教师策略非常接近的性能，因此成为了一种被广泛研究的方法 [10], [14],[15]。与师生两阶段训练方法不同，也有许多方法仅需要单阶段训练过程。DreamWaQ [9] 提出了使用变分自编码器（VAE）来估计状态的隐式表示。Wang等人 [16] 将这种估计方法应用于双足机器人，并评估了估计状态对机器人运动性能的影响。Fu等人 [17] 提出在训练过程中融入编码本体感觉信息的特权学习，并将该方法应用于移动操作任务。Wang等人 [18] 提出了一种同时训练师生网络的方法，该方法在最小化对教师策略影响的同时，显著提升了学生策略的性能。然而，在缺乏外部感知的情况下，克服地形障碍对于腿式机器人，尤其是双足机器人来说，仍然是一个重大挑战。因此，当前对双足机器人的研究主要集中在平坦地形环境上。

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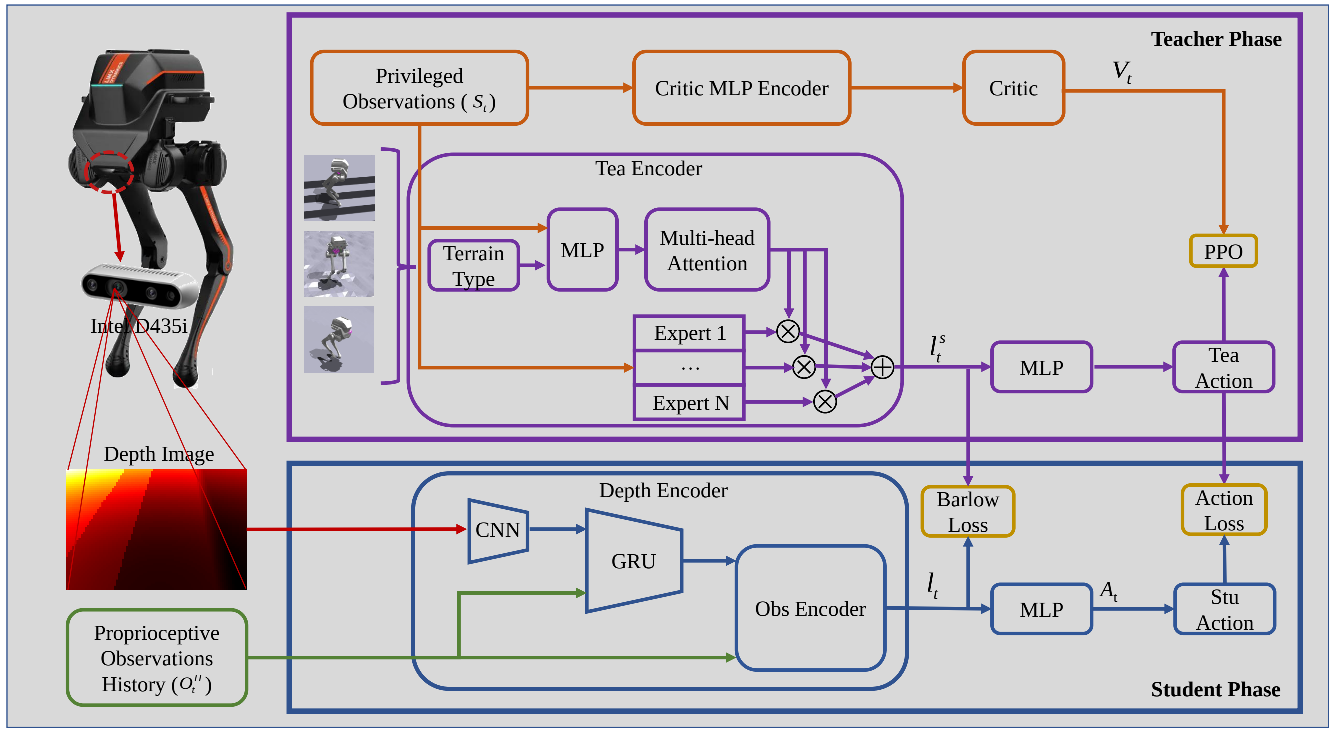


Fig. 2. Training Overview. In the first phase, the network inputs include proprioceptive information, the height map within a one-meter radius around the robot, and accurate physical parameters. Additionally, terrain classification and privileged information are utilized to adjust the weighting of the the mixture of expert, facilitating the training of a reliable teacher policy. In the second phase, the input information consists solely of the current depth images and proprioceptive data. The student policy is trained by aligning the encoded representations of both the student and teacher policies using Barlow Twins, while simultaneously employing supervised learning to guide the student policy to imitate the output of the teacher policy.

The incorporation of external perception aids in overcoming terrain challenges; however, it also increases the difficulty of training legged robots. Despite this, recent advancements have led to the development of several promising solutions [19]–[26]. Reinforcement learning for legged robots with external perception is similarly categorized into teacher-student policies and single-phase training approaches. In the single-phase training framework, Luo et al. [21] extended DreamWaQ [9] by incorporating depth images and employing a VAE-based method to implicitly estimate the surrounding height map of the robot. Meanwhile, Yu et al. [22] proposed a method that utilizes depth images and historical observations to explicitly estimate the height map and body state, enabling quadrupedal robots to overcome challenging terrains such as stairs, steep slopes, and large gaps. However, the single-phase training approach suffers from two major limitations: the significant difficulty of visual perception acquisition in Isaac Gym [27] and the challenge of directly learning the environmental map. Consequently, current research on reinforcement learning for legged robots with external perception predominantly focuses on the teacher-student policy framework. [19] was among the first to propose using online-generated height maps as external input for enabling the student policy to imitate the teacher policy's behavior, achieving favorable experimental results. [24] introduced the direct use of depth images as external input, eliminating the need for online height map generation. [25] employed waypoints as tracking targets to enhance locomotion performance. [26] applied parkour techniques to humanoid robots, successfully enabling them to traverse higher platforms and gaps.

However, the limitation of this two-phase training paradigm lies in the fact that the student policy cannot fully imitate the teacher policy's behavior due to discrepancies in the input information, which may even affect the student policy itself. Moreover, since most current training methods treat different terrains as a single task and employ the same policy network, they overlook potential conflicts that may arise under varying terrain conditions, potentially leading to a degradation in performance.

In this paper, we propose a novel the mixture of experts teacher-student reinforcement learning strategy, which improves the performance of teacher-student policies based on visual inputs through a simple approach. Inspired by [28] and [29], we introduce a mixture of experts model alongside Barlow Twins. By integrating terrain selection strategies with the teacher policy, our method outperforms traditional teacher-student models. Additionally, our approach focuses on aligning the input information between the teacher and student, rather than forcing them to be identical, thereby improving the student's ability to navigate diverse terrains. We deployed our algorithm on the Limx Dynamic P1 bipedal robot and tested it across a variety of terrains.

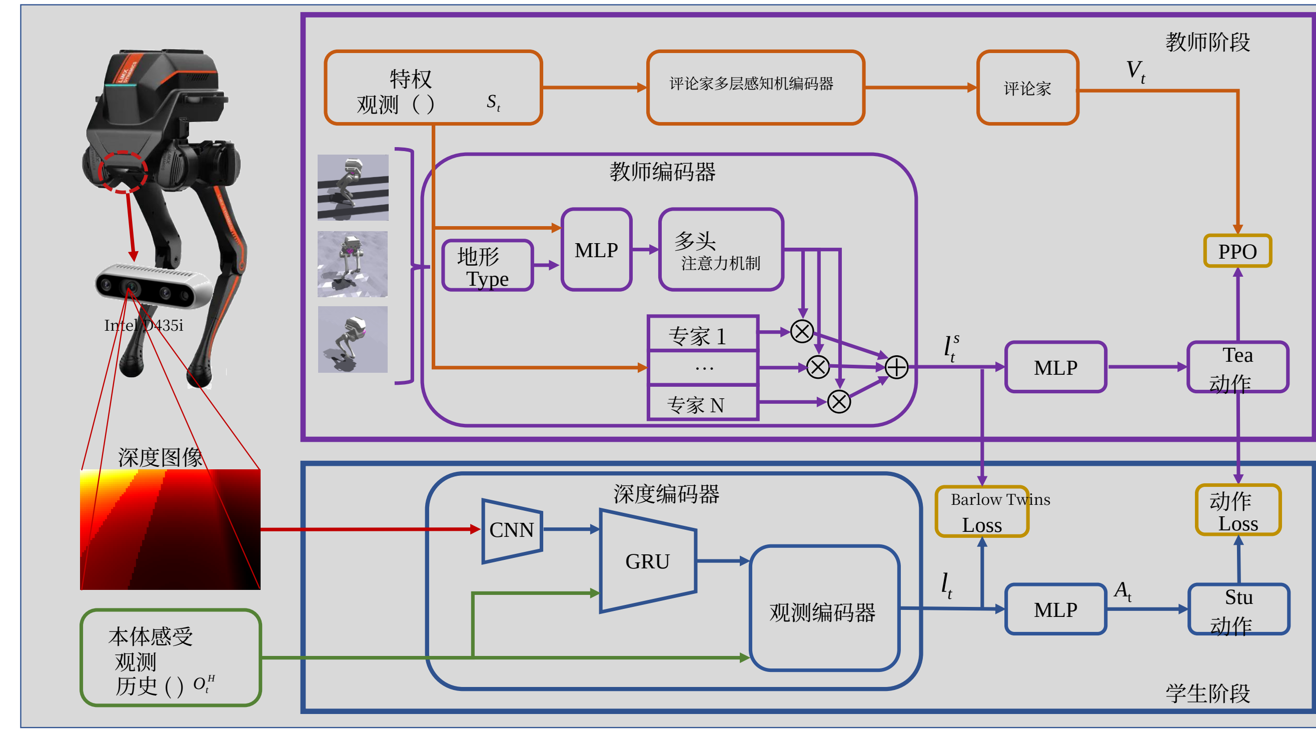


图 2. 训练概览。在第一阶段，网络输入包括本体感觉信息、机器人周围一米半径内的高度图以及准确的物理参数。此外，利用地形分类和特权信息来调整专家混合的权重，从而促进可靠教师策略的训练。在第二阶段，输入信息仅包含当前的深度图像和本体感觉数据。学生策略的训练通过使用 Barlow Twins 对齐学生和教师策略的编码表示，同时采用监督学习引导学生策略模仿教师策略的输出。

引入外部感知有助于克服地形挑战，但同时也增加了训练腿式机器人的难度。尽管如此，近期的进展已经催生了几种有前景的解决方案[19]–[26]。具有外部感知的腿式机器人强化学习同样被分为教师-学生策略和单阶段训练方法。在单阶段训练框架中，Luo等人 [21] 扩展了DreamWaQ [9]，通过引入深度图像并采用基于VAE的方法来隐式估计机器人周围的高度图。同时，Yu等人 [22] 提出了一种利用深度图像和历史观测值来显式估计高度图和身体状态的方法，使四足机器人能够克服楼梯、陡坡和大间隙等具有挑战性的地形。然而，单阶段训练方法存在两个主要局限：在Isaac Gym [27] 中获取视觉感知的难度极大，以及直接学习环境地图的挑战。因此，当前关于具有外部感知的腿式机器人强化学习的研究主要集中在教师-学生策略框架上。 [19] 是最早提出使用在线生成的高度图作为外部输入，使学生策略能够模仿教师策略行为的研究者之一，并取得了良好的实验结果。 [24] 引入了直接使用深度图像作为外部输入，无需在线生成高度图。 [25] 采用航点作为跟踪目标以增强运动性能。 [26]将跑酷技术应用于人形机器人，成功使其能够穿越更高的平台和间隙。

然而，这种两阶段训练范式的局限性在于，由于输入信息的差异，学生策略无法完全模仿教师策略的行为，这甚至可能影响学生策略本身。此外，由于当前大多数训练方法将不同地形视为单一任务并采用相同的策略网络，它们忽略了在不同地形条件下可能出现的潜在冲突，从而导致性能下降。

在本文中，我们提出了一种新颖的专家混合教师-学生强化学习策略，通过一种简单的方法提升了基于视觉输入的师生策略的性能。受 [28]和 [29],启发，我们引入了专家混合模型以及Barlow Twins。通过将地形选择策略与教师策略相结合，我们的方法优于传统的师生模型。此外，我们的方法侧重于对齐教师与学生之间的输入信息，而非强制它们完全相同，从而提升了学生在多样化地形中导航的能力。我们将算法部署在Limx Dynamic P1双足机器人上，并在多种地形上进行了测试。

In summary, the contributions of this paper are as follows:

- First, our framework integrates a mixture of expert models with teacher-student policy learning, where the teacher strategy adaptively adjusts its policy based on terrain, providing a novel vision-aided reinforcement learning framework for legged robots.
- Second, compared to previous approaches that directly distill the teacher's policy to the student, we introduce an implicit alignment and redundancy reduction in the observation feature space between the teacher and student prior to distillation, thereby enhancing the student's performance.
- Finally, we propose a smooth foot-trajectory tracking reward function to mitigate the sim-to-real gap, with comprehensive simulation and physical experiments validating the overall efficacy and superiority of our approach.

II. Method

A. Overview

As shown in Fig. 2, our proposed framework consists of three main components: the teacher policy, the student policy, and the mixture of experts model. Due to the high resource consumption and difficulty in effectively learning visual features in single-phase training, we have opted for a two-phase teacher-student strategy. The input to the student policy includes only the data required for policy deployment, whereas the teacher policy's input, in addition to proprioceptive information, incorporates other privileged information available from the simulation environment. The teacher policy is optimized using the Proximal Policy Optimization (PPO) algorithm [30]. Additionally, the mixture of experts model (detailed in Section II-B), which has been shown to enhance performance in multi-task learning, is integrated into the teacher policy to improve performance across multiple terrains. During the student policy's training process, Barlow Twins (explained in Section II-C) is employed to align the encoded feature space, while the student policy is forced to imitate the teacher policy's actions, enabling it to approximate the teacher policy's behavior.

1) Teacher Policy: The input to the teacher policy s_t, includes the proprioceptive observation o_t, the surrounding height map m_t, and the physical parameters o_p obtained from the simulator, such as mass, velocity, and friction coefficients.

$$s_t = < o_t, o_p, m_t > \tag{1}$$

The proprioceptive observation o_t is a 29-dimensional vector, comprising the angular velocity w_t and gravity vector g_t directly obtained from the IMU, joint positions θ_t, joint velocities $\dot{\theta}_t$, the action a_t from the previous time step, user commands c_t, and the reference contact phase p_t of the foot.

$$o_t = < w_t, g_t, \theta_t, \dot{\theta}_t, a_t, c_t, p_t > \tag{2}$$

Here, p_t represents the reference contact phase of the foot, introduced based on [12] as a square wave function with a period of 0.5s, designed to address the balancing challenge in footed robots that lack a sole.

2) Student Policy: The input to the student policy includes the historical proprioceptive observations o_t^H and the depth information d_t processed through convolutional and recurrent neural networks.

3) Value Network: The input s_t to the evaluation network is the same as that of the teacher network. Therefore, the loss function for the teacher policy training is:

$$\mathcal{L} = \mathcal{L}^{ppo,s} + \mathcal{L}^{value} \tag{3}$$

4) Action Space: The output action a_t, is a 6-dimensional vector, corresponding to the 6 joints of the bipedal robot, representing the deviation between the target positions and the default joint positions. Accordingly, the PD control for the joints is defined as follows:

$$\tau = k_p \times (a_t - \theta_d) - k_d \times \dot{\theta} \tag{4}$$

B. Mixture of Experts

To achieve locomotion control across various terrains, we adopted a curriculum training approach similar to that in [13], categorizing terrains into several types, such as flat ground, slopes, rough terrain, and stairs, as shown in Fig. 3. Each terrain type progressively increases in difficulty as the robot's performance improves during training, such as by increasing the slope angle, staircase height, or the roughness of the ground surface. However, training on different terrains may affect the performance of the policy across these terrain types. Inspired by [28], we introduced a mixture of experts model during the training of the teacher policy. We select the number of experts to be 4. The encoder for the terrain categories and observation inputs is a three-layer fully connected network. After applying multi-head attention and Softmax, the outputs are multiplied by the expert mixture model to obtain the final teacher policy encoding. Therefore, the selection mechanism of the mixture of experts is represented as follows:

$$a_1, a_2, a_3, a_4 = Softmax(W(g(s_t, t_p))) \tag{5}$$

Here, t_p represents the terrain category, a_i denotes the expert weights, W represents the attention, and g refers to the fully connected network.

C. Barlow Twins

During the student policy training phase, as proposed in [19], the output of the teacher policy supervises the output of the student policy, while the privileged information from the teacher policy further supervises the encoded information of the student policy to facilitate the imitation of the teacher's behavior. We adopt a similar approach; however, instead of directly forcing the student's encoded information to match the teacher's, we utilize the Barlow Twins [29] method to align the feature spaces of the student and teacher encodings, rather than enforcing their exact proximity. Therefore, the loss function for the student training process can be expressed as:

$$\mathcal{L} = MSE(\hat{a}_t, a_t) + \sum_i (1 - C_{ii})^2 + \lambda \sum_i \sum_{i \neq j} (1 - C_{ij})^2 \tag{6}$$

I总之， h本文的贡献 ibuti如下 f thi该论文的研究内容如下: follows:

- 首先，我们的框架将专家混合模型与师生策略学习相结合，其中教师策略根据地形的不同自适应调整其策略，为腿式机器人提供了一种新颖的视觉辅助强化学习框架。
- 其次，与以往直接将教师策略蒸馏到学生策略的方法不同，我们在蒸馏之前引入了教师与学生之间观测特征空间的隐式对齐与冗余减少，从而提升了学生策略的性能。
- 最后，我们提出了一种平滑足部轨迹跟踪奖励函数，以缩小仿真到现实差距，并通过全面的仿真和物理实验验证了我们方法的整体有效性和优越性。

II. 方法

A. 概述

如图2所示，我们提出的框架包含三个主要组成部分：教师策略、学生策略和专家混合模型。由于单阶段训练中资源消耗高且难以有效学习视觉特征，我们选择采用两阶段师生策略。学生策略的输入仅包含策略部署所需的数据，而教师策略的输入除了本体感觉信息外，还包含仿真环境中可用的其他特权信息。教师策略使用近端策略优化（PPO）算法 [30]进行优化。此外，专家混合模型（详见第II-B节）已被证明能提升多任务学习性能，因此被集成到教师策略中，以改善其在多种地形上的表现。在学生策略的训练过程中，采用Barlow Twins（详见第II-C节）对齐编码特征空间，同时强制学生策略模仿教师策略的动作，使其能够逼近教师策略的行为。

1) 教师策略：教师策略 s_t 的输入包括本体感觉观测 o_t、周围高度图 m_t，以及从模拟器中获取的物理参数 o_p ，例如质量、速度和摩擦系数。

$$s_t = < o_t, o_p, m_t > \tag{1}$$

本体感觉观测 o_t 是一个 29 维向量，包含直接从惯性测量单元获取的角速度 w_t 和重力向量 g_t、关节位置 θ_t、关节˙速度 θ_t、上一时间步的动作 a_t 、用户指令 c_t、以及足部的参考接触相位 p_t 。

$$o_t = < w_t, g_t, \theta_t, \dot{\theta}_t, a_t, c_t, p_t > \tag{2}$$

这里， p_t 表示足部的参考接触相位，基于 [12] 引入，作为周期为0.5秒的方波函数，旨在解决缺乏脚掌的有足机器人所面临的平衡挑战。

2) 学生策略：学生策略的输入包括历史本体感觉观测 o_t^H以及通过卷积神经网络和循环神经网络处理的深度信息 d_t 。

3) 价值网络：评估网络 s_t 的输入与教师网络相同。因此，教师策略训练的损失函数为：

$$\mathcal{L} = \mathcal{L}^{ppo,s} + \mathcal{L}^{value} \tag{3}$$

4) 动作空间：输出动作 a_t 是一个 6 维向量，对应双足机器人的 6 个关节，表示目标位置与默认关节位置之间的偏差。相应地，关节的 PD控制定义如下：

$$\tau = k_p \times (a_t - \theta_d) - k_d \times \dot{\theta} \tag{4}$$

B. 专家混合模型

为了实现跨多种地形的运动控制，我们采用了类似于[13]，的课程训练方法，将地形分为几种类型，如平坦地面、斜坡、崎岖地形和楼梯，如图3所示。随着训练过程中机器人性能的提升，每种地形类型的难度会逐步增加，例如通过增大坡度角、楼梯高度或地面粗糙度。然而，在不同地形上训练可能会影响策略在这些地形类型上的表现。受 [28]，启发，我们在教师策略的训练过程中引入了专家混合模型。我们将专家数量设为4个。地形类别和观测输入的编码器采用三层全连接网络。经过多头注意力和Softmax处理后，输出结果与专家混合模型相乘，从而获得最终的教师策略编码。因此，专家混合模型的选择机制表示如下：

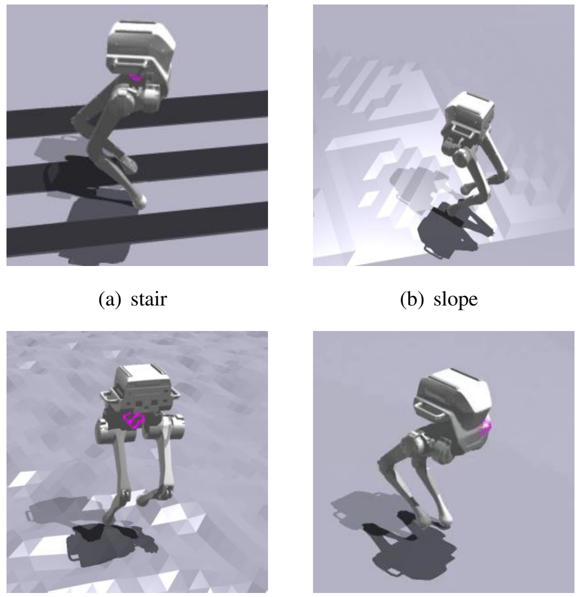
$$a_1, a_2, a_3, a_4 = Softmax(W(g(s_t, t_p))) \tag{5}$$

这里， t_p 表示地形类别， a_i 表示专家权重， W 表示注意力机制， g 指代全连接网络。

C. Barlow Twins

在学生策略训练阶段，如[19],所述，教师策略的输出监督学生策略的输出，同时教师策略中的特权信息进一步监督学生策略的编码信息，以促进对教师行为的模仿。我们采用了类似的方法；然而，我们并非直接强制学生的编码信息与教师的编码信息匹配，而是利用Barlow Twins [29] 方法来对齐学生编码与教师编码的特征空间，而非强制它们精确接近。因此，学生训练过程的损失函数可以表示为：

$$\mathcal{L} = MSE(\hat{a}_t, a_t) + \sum_i (1 - C_{ii})^2 + \lambda \sum_i \sum_{i \neq j} (1 - C_{ij})^2 \tag{6}$$



(a) stair (b) slope

(c) rough (d) plane

Fig. 3. Terrain type

TABLE I

REWARD FUNCTION ELEMENTS

Reward	Equation	Weight
Lin. velocity tracking	$exp(-4\lVert v_{xy}^{cmd} - v_{xy}\rVert^2)$	7.5
Ang. velocity tracking	$exp(-4\lVert w_z^{cmd} - w_z\rVert^2)$	4.0
Lin. velocity (z)	v_z^2	-0.5
Ang. velocity (xy)	w_{xy}^2	-0.06
Orientation	g^2	-6.0
Base Height	$\lVert h^{target} - h\rVert$	-10.0
Joint acceleration	$\ddot{q}^2$	-2.5 × 10^{-7}
Action rate	$\lVert a_t - a_{t-1}\rVert_2^2$	-0.01
Feet air time	$\lVert max(t_{air}, t_{min})\rVert$	60
Joint torque	$\lVert \tau\rVert_2^2$	-2.5 × 10^{-5}
Feet contact force	$\lVert f_{feet} - f_{max}\rVert_2^2$	-0.01
Feet distance	$\lVert p_{left} - p_{right}\rVert_2^2$	-100
Unbalance feet height	$\lVert h_{left}^{max} - h_{right}^{max}\rVert$	-60
Unbalance feet air time	$\lVert t_{left}^{max} - t_{right}^{max}\rVert$	-300
Collision	$n_{collisin}\rVert$	-30
Joint torque limitation	$max(\tau - \tau_{limit}, 0)$	-0.1
Target feet height	$\lVert h_{feet}^{target} - h_{feet}\rVert$	-6.0
Feet position (x)	$\lVert (p_x^{left} - p_x^{right})\rVert$	-3.0
Hip Position	$\lVert q_{hip} - q_{hip}^{default}\rVert$	-1.5
Feet Contact Number	r^{fn}	2.5
Lin. velocity smooth	$\lVert \dot{v}_{xy}^t - \dot{v}_{xy}^{t-1}\rVert$	-0.05
Survival	$t_{survival}$	-0.05
Feet Velocity	r^{fv}	-0.5

Here, $\hat{a}_t$ represents the output of the teacher policy, $\hat{a}_t$ denotes the output of the student policy, and C refers to the cross-correlation matrix between the student encoding Z_a and the teacher encoding Z_b.

III. TRAIN DETAILS

A. Reward Function Design

To achieve stable locomotion for the bipedal robot, we designed the reward function as shown in Table I, with reference to [18]. Here, f_{feet} represents the foot contact force, h denotes the body height, t_{air} indicates the foot airtime, p_{left} and p_{right} represent the positions of the left and right feet in the body coordinate system, h_{feet}^{max} refers to the maximum height of feet lift, $n_{collision}$ denotes the number of collision points, τ represents the joint torques, and q_{hip} indicates the position of the hip joint. In order to facilitate smooth foot landing during the training process, we impose a penalty on the foot contact force. However, due to the inherent inaccuracies in the foot contact force measurements within the simulator, we supplement the force penalty with an additional penalty on the vertical velocity of the foot when it is within a certain distance from the ground, in order to achieve the same objective. The expression for this penalty is as follows:

$$r^{fv} = v_z^{feet} \times (h_{feet} < 0.05) \tag{7}$$

Due to the lack of a footplate structure in point-foot bipedal robots, maintaining balance becomes challenging. Therefore, we introduce the foot reference phase as described in [12], and penalize the robot's foot phase when it deviates from the reference phase. The reward formula is as follows:

$$r^{fn} = \sum_i^N (1 \times (C_i = S_i) + (-1) \times (C_i \neq S_i)) \tag{8}$$

$$C_i = \begin{cases} 0 & \text{if } sin(\frac{2\pi t}{T} + \theta) \geq 0 \\ 1 & \text{if } sin(\frac{2\pi t}{T} + \theta) < 0 \end{cases} \tag{9}$$

$$\theta = \begin{cases} 0 & \text{if } i = left \\ \pi & \text{if } i = right \end{cases} \tag{10}$$

Here, N represents the number of feet, C_i denotes the reference phase for the i-th foot, and S_i represents the actual phase of the i-th foot, T=0.5s denotes the period.

Due to the tendency of bipedal robots to continuously raise the foot height during training in an attempt to traverse various terrains, this behavior is detrimental to maintaining balance. Therefore, to constrain the foot height during training and ensure the smoothness of the foot trajectory, we have designed the following reference foot trajectory:

$$h_{feet,i}^{target} = \frac{1}{2} max(h_{feet}^{max} \times (1 - cos(\frac{4\pi t}{T}), 0) \times \neg C_i \tag{11}$$

Here, h_{feet}^{max} represents the maximum foot height.

B. Environment Setup

We employed NVIDIA Isaac Gym [27] as the simulation environment to train 8192 robots in parallel, with both the teacher and student policies utilizing 8192 robots. The teacher policy converged after approximately 4000 training episodes, requiring around 3 hours on an RTX 3090 GPU. During the training of the student policy, the substantial memory consumption and longer acquisition times for visual

(a) 楼梯 (b) 斜坡

(c) 粗糙地形 (d) 平面

图 3. 地形类型

表I 奖励函数元素

奖励	公式	权重
线速度跟踪	$exp(-4\lVert v_{xy}^{cmd} - v_{xy}\rVert^2)$	7.5
角速度跟踪	$exp(-4\lVert w_z^{cmd} - w_z\rVert^2)$	4.0
线速度（z轴）	v_z^2	-0.5
角速度（xy平面）	w_{xy}^2	-0.06
朝向	g^2	-6.0
基座高度	$\lVert h^{target} - h\rVert$	-10.0
关节加速度	$\ddot{q}^2$	-2.5 × 10^{-7}
动作频率	$\lVert a_t - a_{t-1}\rVert_2^2$	-0.01
足部腾空时间	$\lVert max(t_{air}, t_{min})\rVert$	60
关节力矩	$\lVert \tau\rVert_2^2$	-2.5 × 10^{-5}
足底接触力	$\lVert f_{feet} - f_{max}\rVert_2^2$	-0.01
足间距	$\lVert p_{left} - p_{right}\rVert_2^2$	-100
足部不平衡高度	$\lVert h_{left}^{max} - h_{right}^{max}\rVert$	-60
足部不平衡腾空时间	$\lVert t_{left}^{max} - t_{right}^{max}\rVert$	-300
碰撞	$n_{collisin}\rVert$	-30
关节力矩限制	$max(\tau - \tau_{limit}, 0)$	-0.1
目标足高	$\lVert h_{feet}^{target} - h_{feet}\rVert$	-6.0
足部位置（x）	$\lVert (p_x^{left} - p_x^{right})\rVert$	-3.0
髋部位置	$\lVert q_{hip} - q_{hip}^{default}\rVert$	-1.5
足部接触数量	r^{fn}	2.5
线速度平滑	$\lVert \dot{v}_{xy}^t - \dot{v}_{xy}^{t-1}\rVert$	-0.05
生存	$t_{survival}$	-0.05
足部速度	r^{fv}	-0.5

这里，$\hat{a}_t$ 表示教师策略的输出，$\hat{a}_t$表示学生策略的输出， C 指的是学生编码Z_a 与教师编码 Z_b之间的互相关矩阵。

III. 训练细节

A. 奖励函数设计

为了实现双足机器人的稳定运动，我们参考 [18]设计了如表I所示的奖励函数。其中， f_{feet} 表示足底接触力， h 表示身体高度， t_{air} 表示足部腾空时间， p_{left} 和 p_{right} 表示左右足部在身体坐标系中的位置， h_{feet}^{max} 表示足部抬升最大高度， $n_{collision}$ 表示碰撞点数量， τ 表示关节力矩， q_{hip}表示髋关节位置。为了在训练过程中促进足部平稳着地，我们对足底接触力施加了惩罚。然而，由于模拟器中足底接触力测量存在固有的不准确性，我们在足部距离地面一定距离时，对其垂直速度施加额外的惩罚，以补充力惩罚，从而实现相同目标。该惩罚的表达式如下：

$$r^{fv} = v_z^{feet} \times (h_{feet} < 0.05) \tag{7}$$

由于点足双足机器人缺乏足板结构，保持平衡变得具有挑战性。因此，我们引入 [12], 中描述的足部参考相位，并在机器人的足部相位偏离参考相位时对其进行惩罚。

奖励公式如下：

$$r^{fn} = \sum_i^N (1 \times (C_i = S_i) + (-1) \times (C_i \neq S_i)) \tag{8}$$

$$C_i = \begin{cases} 0 & \text{if } sin(\frac{2\pi t}{T} + \theta) \geq 0 \\ 1 & \text{if } sin(\frac{2\pi t}{T} + \theta) < 0 \end{cases} \tag{9}$$

$$\theta = \begin{cases} 0 & \text{if } i = left \\ \pi & \text{if } i = right \end{cases} \tag{10}$$

这里， N 表示足部数量， C_i 表示第i个足部的参考相位， S_i 表示第i个足部的实际相位，T=0.5s表示周期。

由于双足机器人在训练过程中倾向于持续抬高足部高度以试图穿越各种地形，这种行为不利于保持平衡。因此，为了在训练中约束足部高度并确保足部轨迹的平滑性，我们设计了以下参考足部轨迹：

$$h_{feet,i}^{target} = \frac{1}{2} max(h_{feet}^{max} \times (1 - cos(\frac{4\pi t}{T}), 0) \times \neg C_i \tag{11}$$

这里， h_{feet}^{max} 表示最大足部高度。

B. 环境设置

我们采用 NVIDIA Isaac Gym [27]作为仿真环境，并行训练了8192个机器人，教师策略和学生策略均使用了8192个机器人。教师策略在大约4000个训练回合后收敛，在RTX 3090 GPU上耗时约3小时。在学生策略训练过程中，Isaac Gym中视觉信息的大量内存消耗和较长的获取时间显著增加了训练时间。

TABLE II
PARAMETERS SETUP

Parameter	Value	Unit
Base height	0.66	m
Feet height	0.03	m
Action scale	0.25	-
Decimation	4	-
Kp	40.0	N/rad
Kd	2.0	$N \cdot s/rad$
Lin. velocity range (x)	[-0.5, 0.5]	m/s
Lin. velocity range (y)	[-0.2, 0.2]	m/s
Ang. velocity range (y)	[-0.5, 0.5]	rad/s

TABLE III
DOMAIN RANDOMIZATION

Parameter	Range	Unit
Payload mass	[-1, 2]	Kg
Center of mass shift	[-3,3]×[-2,2]×[-3,3]	cm
K_p Factor	[0.8,1.2]	N/rad
K_d Factor	[0.8,1.2]	N·s/rad
Friction	[0.2,1.6]	-
Restitution	[0.0,1.0]	-
Step delay	[0, 15]	ms
Push vel (xy)	0.5	m/s
Camera position	[-5,5]×[-10,10]	mm
Camera angle (pitch)	[-1,1]	deg
Camera Fov	[86,88]	deg

information in Isaac Gym significantly increased the training time. Specifically, training the student policy for 4000 episodes, building on the teacher policy, took approximately 120 hours. To address this, we utilized NVIDIA Warp [31] to store terrain mesh data during terrain initialization. When acquiring depth images, we adopted a ray-casting approach based on the robot's camera pose and physical properties, rather than relying solely on Isaac Gym's visual data acquisition. This modification, leveraging NVIDIA Warp for depth images retrieval, reduced the training time for the student policy to approximately 16 hours.During the training process, certain training parameters of the robot, such as PD parameters, target base height, foot end height, and others, are configured as shown in Table II.

C. Domain Randomization

To enhance the model's generalization ability and reduce the simulation-to-reality gap, thereby facilitating the deployment of models trained in simulation on real robots, we applied threshold randomization within a specified range to certain physical parameters of the robot in the simulation. These parameters include PD controller gains, friction coefficients, joint masses, camera parameters, and others. The specific settings for the threshold randomization are presented in Table III.

IV. EXPERIMENT

A. Simulation Experiments

To evaluate the effectiveness of the proposed method, we performed a comparison with several alternative approaches

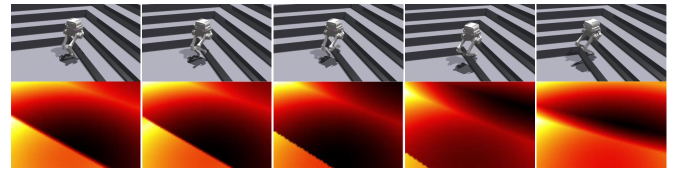


Fig. 4. The motion timing diagram of a bipedal robot with visual assistance in simulation. The lower section of the diagram illustrates the depth images obtained through NVIDIA Warp and ray casting during the robot's movement.

within the simulation environment. All methods were trained under the identical setup described in Section III-B, with the reward function held constant. The comparison was conducted by only modifying the network architecture and loss functions. The detailed descriptions of these methods are as follows:

- Blind: The input consists solely of proprioceptive information, and training is conducted using the Proximal Policy Optimization (PPO) algorithm.
- Teacher-Student policy(TS): The two-phase teacher-student strategy utilizes depth images and proprioceptive information as inputs. The teacher policy is trained first, followed by the training of the student policy. A detailed description of this approach can be found in [25].
- PIE: The single-phase strategy simultaneously trains the model while explicitly or implicitly predicting information such as velocity, body height, and height maps, as referenced in [21].

To ensure a fair comparison, for the single-phase training method, we selected data from a total of 8000 iterations. For the two-phase training method, we chose 4000 iterations from each phase, resulting in a total of 8000 iterations, with 4000 iterations from phase one and 4000 iterations from phase two.

We selected the average reward and terrain level as evaluation metrics during the training process. The average reward serves as an indicator of the quality of the training task, while the terrain level assesses the robot's ability to overcome various terrains. The average reward and terrain level curves during the training processes of our method and several other approaches are presented in Fig. 5 and Fig. 6, respectively.

Fig. 5 illustrates that, during the training process, the reward of our teacher policy exceeds that of the TS-teacher. The reward of the student policy is influenced by the performance of the teacher policy, resulting in a higher reward for our student policy compared to TS-student. In contrast, PIE focuses more on the velocity term due to its prediction of variables such as speed, thereby assigning higher weight to the velocity component in the reward function. Consequently, the final reward of our method closely approximates that of PIE. Fig. 6 illustrates that the terrain level of our student policy surpasses that of TS-student. Although PIE incorporates terrain prediction, it still struggles to effectively overcome challenging terrains. Consequently, the terrain level of our

表II 参数设置

参数	数值	Unit
基座高度	0.66	m
脚部高度	0.03	m
动作尺度	0.25	-
降采样	4	-
Kp	40.0	N/rad
Kd	2.0	$N \cdot s/rad$
线速度范围 (x)	[-0.5, 0.5]	m/s
线速度范围 (y)	[-0.2, 0.2]	m/s
角速度范围 (y)	[-0.5, 0.5]	rad/s

表三 域随机化

参数	范围	Unit
有效载荷质量	[-1, 2]	Kg
质心偏移	[-3,3]×[-2,2]×[-3,3]	cm
K_p 因子	[0.8,1.2]	N/rad
K_d 因子	[0.8,1.2]	N·s/rad
摩擦	[0.2,1.6]	-
恢复系数	[0.0,1.0]	-
步进延迟	[0, 15]	ms
推送速度（xy）	0.5	m/s
相机位置	[-5,5]×[-10,10]	mm
相机角度（俯仰）	[-1,1]	deg
相机视场角	[86,88]	deg

具体而言，在教师策略基础上训练学生策略4000个回合，耗时约120小时。为解决此问题，我们在地形初始化阶段利用 NVIDIA Warp [31]存储地形网格数据。在获取深度图像时，我们采用基于机器人相机位姿和物理属性的光线投射方法，而非仅依赖 Isaac Gym 的视觉数据获取。这一改进利用 NVIDIA Warp 进行深度图像检索，将学生策略的训练时间缩短至约16小时。在训练过程中，机器人的某些训练参数，如PD参数、目标基座高度、足端高度等，配置如表II所示。

C. 域随机化

为了增强模型的泛化能力并缩小仿真到现实的差距，从而便于将在仿真中训练的模型部署到真实机器人上，我们对仿真中机器人的某些物理参数在指定范围内进行了阈值随机化。这些参数包括PD控制器增益、摩擦系数、关节质量、相机参数等。阈值随机化的具体设置见表III。

IV. 实验

A. 仿真实验

为了评估所提出方法的有效性，我们与几种替代方法进行了比较

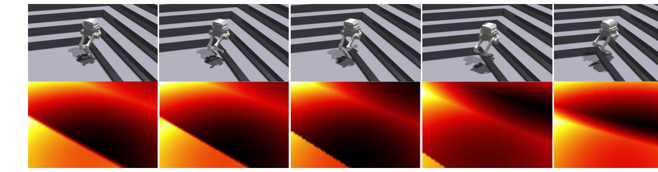


图4. 具有视觉辅助的双足机器人在仿真中的运动时序图。图的下半部分展示了机器人运动过程中通过NVIDIA Warp和光线投射获得的深度图像。

在仿真环境中进行。所有方法均在第三节B部分描述的相同设置下进行训练，并保持奖励函数不变。比较仅通过修改网络架构和损失函数来进行。这些方法的详细描述如下：

- 盲策略：输入仅包含本体感觉信息，并使用近端策略优化（PPO）算法进行训练。
- 教师-学生策略（TS）：两阶段教师-学生策略利用深度图像和本体感觉信息作为输入。首先训练教师策略，随后训练学生策略。该方法的详细描述见[25]。
- PIE：单阶段策略在训练模型的同时，显式或隐式地预测速度、身体高度和高度图等信息，详见 [21]。

为确保公平比较，对于单阶段训练方法，我们选取了总共8000次迭代的数据。对于两阶段训练方法，我们从每个阶段选取了4000次迭代，总计8000次迭代，其中第一阶段4000次迭代，第二阶段4000次迭代。

我们选取平均奖励和地形等级作为训练过程中的评估指标。平均奖励用于指示训练任务的质量，而地形等级则评估机器人克服各种地形的能力。我们的方法及其他几种方法在训练过程中的平均奖励和地形等级曲线分别如图5和图6所示。

图5展示了在训练过程中，我们的教师策略的奖励超过了TS-教师。学生策略的奖励受到教师策略表现的影响，因此我们的学生策略相比TS-学生获得了更高的奖励。相比之下，PIE由于预测了速度等变量，更关注速度项，从而在奖励函数中为速度分量分配了更高的权重。因此，我们方法的最终奖励与PIE非常接近。图6显示，我们的学生策略的地形等级超过了TS-学生。尽管PIE引入了地形预测，但它仍然难以有效克服具有挑战性的地形。因此，我们的

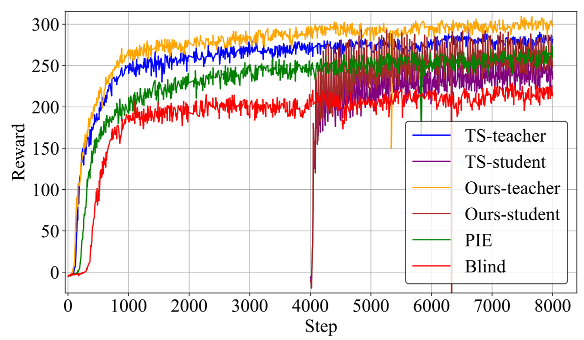


Fig. 5. The reward curves during the training process of our method and several other approaches.

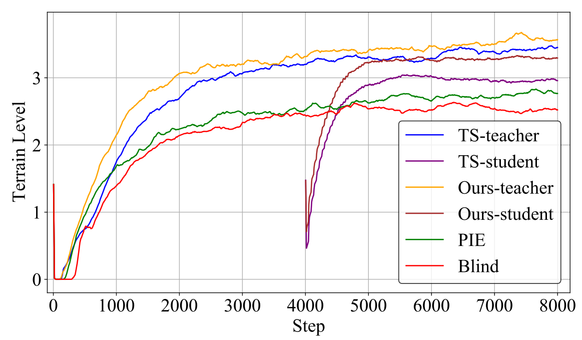


Fig. 6. The terrain level curves during the training process of our method and several other approaches.

student policy exceeds that of PIE, while PIE performs better than the blind policy.

To validate the effect of the alignment module (Barlow Twins) on the student policy, we compared the terrain level of the TS strategy, the TS strategy with the alignment module, and our proposed method. The results, as shown in Fig. 7, demonstrate that the terrain level of the strategy with the alignment module exceeds that of the original TS strategy, yet still falls short of the performance achieved by our method.

TABLE IV
EVALUATE RESULT

Method	Vel. Error	Height Error	Survival Time
Blind	0.929	0.076	84.675
PIE	0.546	0.084	87.111
TS	0.673	0.097	88.601
Ours	**0.535**	**0.071**	88.485
Ours w/o barlow	0.554	0.079	**88.652**
Ours w/o Moe	0.593	0.093	87.011

In the simulation, we deployed 128 robots across several terrains with fixed proportions to evaluate the performance of different strategies in terms of speed tracking, average survival time within 100 seconds, and height tracking error, which are considered fundamental performance metrics. The results are presented in Table IV. In the simulation experi-

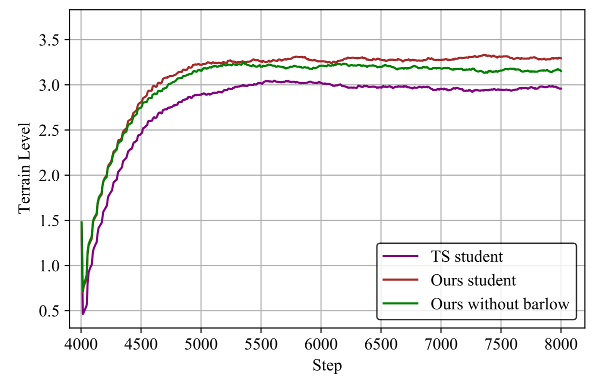


Fig. 7. The terrain level curves during the training process of the TS policy, incorporating the Barlow Twins method, and our proposed method.

Fig. 8. The bipedal robot maneuvers and turns on grassland.

ments, our framework outperforms both the teacher-student policy and the PIE strategy in terms of basic user command tracking and survival time. In the case of w/o Moes, our strategy exhibits slightly lower velocity tracking performance compared to PIE, which is attributed to PIE's real-time estimation of the robot's state. However, in the case of w/o Barlow Twins, our strategy shows a significant improvement over the TS policy, although its velocity tracking error still slightly exceeds that of the PIE strategy.

B. Real-World Experiments

We deployed our strategy on the Limx Dynamics P1 robot, which features a 6-DOF body and is equipped with an IMU and a D435i sensor. We tested our strategy across various

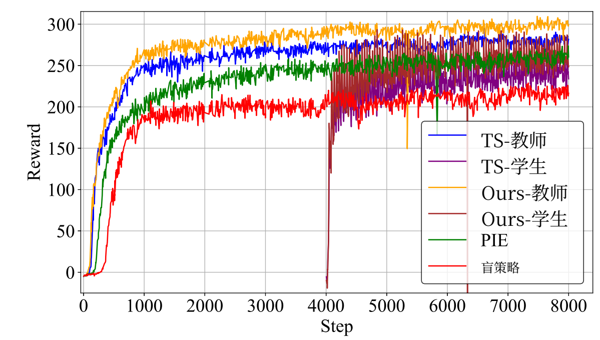


图5. 我们的方法及其他几种方法在训练过程中的奖励曲线。

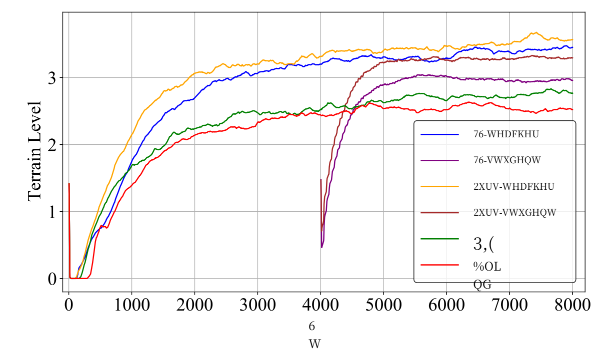


图6. 我们的方法及其他几种方法在训练过程中的地形等级曲线。

学生策略的地形等级超过了PIE，而PIE的表现优于盲策略。

为了验证对齐模块（Barlow Twins）对学生策略的影响，我们比较了TS策略、带有对齐模块的TS策略以及我们提出的方法的地形等级。结果如图7所示，表明带有对齐模块的策略的地形等级超过了原始TS策略，但仍未达到我们方法所实现的性能。

表IV 评估结果

方法	速度误差	高度误差	生存时间
盲策略	0.929	0.076	84.675
PIE	0.546	0.084	87.111
TS	0.673	0.097	88.601
Ours	**0.535**	**0.071**	88.485
我们的方法（无Barlow）	0.554	0.079	**88.652**
我们的方法（无MoE）	0.593	0.093	87.011

在仿真中，我们在多个地形上以固定比例部署了128个机器人，以评估不同策略在速度跟踪、100秒内平均生存时间以及高度跟踪误差方面的表现，这些被视为基础性能指标。结果展示在表IV中。在仿真实验

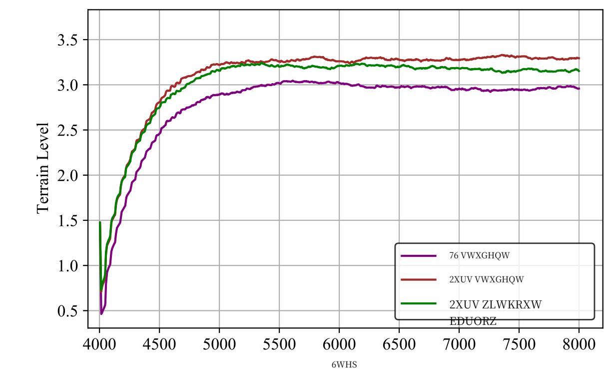


图7. TS策略在训练过程中的地形等级曲线，包含Barlow Twins方法以及我们提出的方法。

图8. 双足机器人在草地上进行机动和转向。

中，我们的框架在基本用户指令跟踪和生存时间方面均优于教师-学生策略和PIE策略。在无Moes的情况下，我们的策略在速度跟踪性能上略低于PIE，这归因于PIE对机器人状态的实时估计。然而，在无Barlow Twins的情况下，我们的策略相比TS策略有显著提升，尽管其速度跟踪误差仍略高于PIE策略。

B. 真实世界实验

我们将策略部署在配备六自由度机身、惯性测量单元和D435i传感器的Limx Dynamics P1机器人上，并在多种地形上进行了测试。

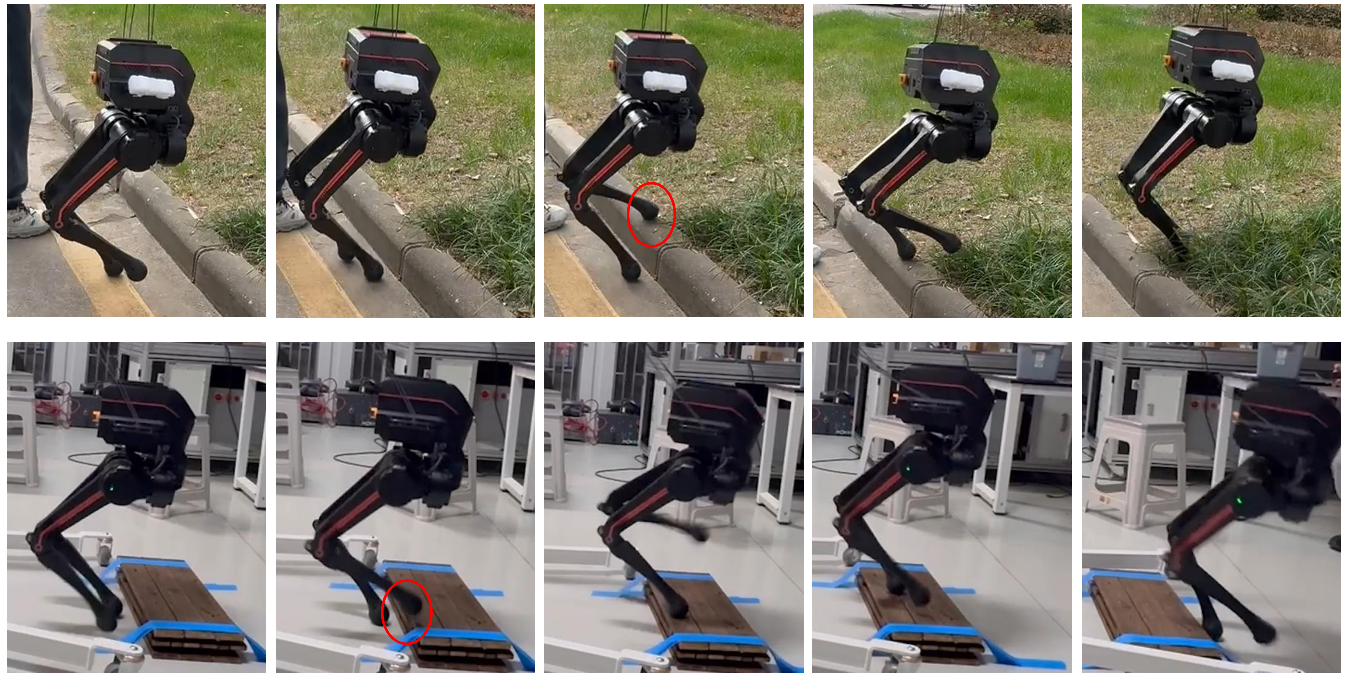


Fig. 9. The bipedal robot adjusts its gait to overcome obstacles upon detecting them. The red circle in the figure indicates the robot autonomously raising its foot height to traverse the terrain.

terrains.

As shown in Fig. 8, we deployed our strategy on grassland. Despite the soft grass not providing the same stable friction and elasticity as a flat surface, the bipedal robot is still able to achieve stable locomotion and turning.

To validate the robot's ability to recognize and overcome obstacles, we evaluated its capacity to traverse stairs and surfaces elevated above the current ground level. As shown in Fig. 9, when the robot detects an obstacle that can be crossed, it increases the foot's lift height to ensure stable clearance. After overcoming the obstacle, the robot lowers the foot height to maintain its stability. This demonstrates that our strategy, supported by vision, enables the robot to effectively navigate over obstacles. As shown in Fig. 10, the strategy that does not use camera depth images as input is highly prone to falling due to unintended contact with unknown terrain.

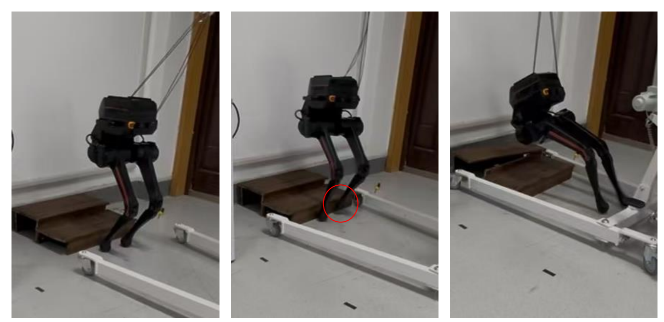


Fig. 10. The robot does not autonomously raise its foot height and falls while blind walking. The red circle in the figure indicates the point where the foot makes contact with the vertical portion of the step.

V. CONCLUSION AND FUTURE WORKS

We propose a novel vision-based reinforcement learning strategy that combines the mixture of experts model, contrastive learning, and teacher-student models, yielding slightly better performance than the teacher-student model alone. We validated the effectiveness of this approach through both simulation comparisons and real-world deployment, demonstrating its capability to traverse various terrains. However, the inherent instability of the chosen bipedal platform, such as difficulty in maintaining a stable speed, posed significant challenges to our experiments. Furthermore, the current work is limited to control and does not address trajectory planning. Therefore, future work will focus on enhancing the stability of the bipedal robot and developing dynamic path planning strategies.

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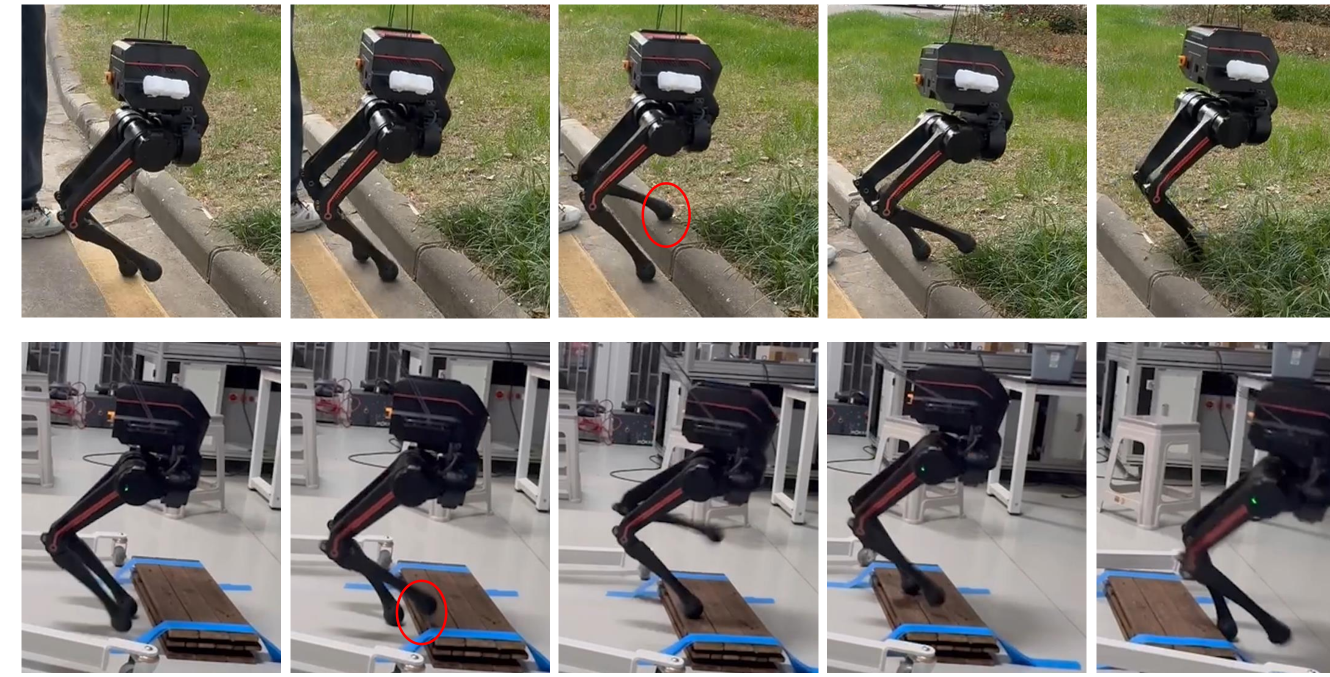

图9. 双足机器人在检测到障碍物后调整步态以跨越障碍。图中红色圆圈标示了机器人自主提升足部高度以穿越地形。

地形。

如图8所示，我们在草地上部署了我们的方法。尽管柔软的草地无法提供与平坦表面相同的稳定摩擦和弹性，但双足机器人仍能实现稳定运动和转向。

为验证机器人识别并跨越障碍物的能力，我们评估了其穿越楼梯及高于当前地面高度的表面。如图9所示，当机器人检测到可跨越的障碍物时，它会增加足部抬升高度以确保稳定通过。跨越障碍后，机器人降低足部高度以维持稳定性。这表明，在视觉支持下，我们的方法使机器人能够有效导航越过障碍物。如图10所示，未使用相机深度图像作为输入的方法极易因与未知地形发生意外接触而摔倒。

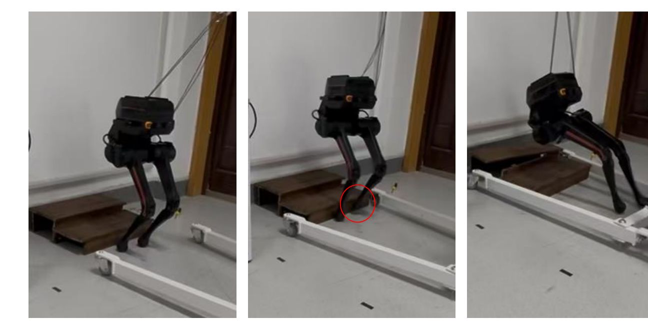

图10. 机器人在盲策略行走时未能自主提升足部高度而摔倒。图中红色圆圈标示了足部与台阶垂直部分发生接触的位置。

V. 结论与未来工作

我们提出了一种新颖的基于视觉的强化学习策略，该策略结合了专家混合模型、对比学习和师生模型，其性能略优于仅使用师生模型。我们通过仿真对比和实际部署验证了该方法的有效性，展示了其穿越各种地形的能力。然而，所选双足平台固有的不稳定性，例如难以保持稳定速度，给我们的实验带来了重大挑战。此外，当前工作仅限于控制，未涉及轨迹规划。因此，未来的工作将集中于增强双足机器人的稳定性，并开发动态路径规划策略。

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